Solar Reflector

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Table of Contents

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Abstract

The sun is a wonderful thing. It is one of the biggest reasons why our planet has life on it in the first place. The sun provides energy in multiple portions of the electromagnetic spectrum, heating up our planet to just the right temperature and even giving us light. In the search for clean energy, the sun has also been a huge help to us all, in that we can now harness its visible energy with solar panels. However, the sun's visible effects only last for roughly half a day. We still receive heat to our planet, which is trapped inside thanks to the atmosphere, but light escapes comparatively easily, hence why nights are dark (and only slightly colder in our area of the world). Half a day's worth of light is still a lot, but as buildings get taller, more and more shade fills the planet. Cloudy days further limit our access to sunlight. Unfortunately, we cannot move the clouds. We cannot make sunlight pass through buildings. What we can do, however, is make the most of the sunlight that is hitting us. We plan to build a device that allows us to re-route light to locations that may not have much of it at all. This device is known as a heliostat, and we have made one that is small enough to fit in one's garden or outside a home. With our system, solar panels could still receive great amounts of light on angled roofs even if they are facing away from the sun at some point of the day. We could keep natural light in our homes for more hours of the day. We could even bring light to gardens that may be stuck in shade, covered by townhouses or trees. Through this research, we found that it is not only possible to build such a system, but we've also come extremely close to getting it done. In doing so, we discovered the importance of torque against gravity in such systems, a situation we combated by using cycloidal drive setups, which gave us 11 times the torque, allowing us to lift our mirror system against gravity. This design has great potential and could help the average person get just a little bit more sunlight in their lives. Large heliostat systems already exist, but no person could attach one of these large heliostats to their homes. Here, we make one that is small.

I. Introduction:

The sun moves. This means that almost everyone on the planet gets to have sunlight every 24 hours. However, the movement of this large source of light also moves shadows. This means that things like gardens and solar panels don't receive sunlight at certain times of the day, even if the sun is up. Take, for example, a small garden in the back of a home. The plants may receive sunlight for half the day, but what about the portion of the time when the sun is on the other side of the home? The entire garden would be left in darkness, yet there's all this untapped sunlight going around the sides of the home, just waiting to be used. If we want to harness the energy of the sun, we need to maximize our access to it. We want to take sunlight from areas that receive it for a large majority of the day, and then reflect it to other places, such as the side of a home.

II. Background:

Shadows are the result of light's path to an object being obstructed by other solids. Since light follows a direct path from the sun to the earth, light usually only travels in one direction. If that path is obstructed, the obstructing object receives the light, and all objects behind receive none. However, there is one way of changing the path of light that we're all familiar with. A mirror reflects light in such a way that the output angle matches the incident angle, just in the opposing direction [3]. Imagine taking a vector and then just flipping the component of that

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vector that points towards the mirror. This makes the redirection of light with mirrors very predictable.

This is the principle that heliostats work under. They calculate the angle at which the mirror must be pointed to keep the sunlight on the target point.

III. Related Work:

There are quite a few solutions to the problem of shadows that already exist. The first, as mentioned before, is the classic heliostat [4]. However, these are extremely large. Nobody would ever use such a large system near their own home, as there would be no way to attach it to the side or stick it into the ground without obstructing much of their own space. If we want to use heliostats, we need to make them smaller. Strangely enough, small heliostats already exist. The Wikoda Sunflower is a commercial product that anyone could buy and place near their own garden in order to maximize the amount of light their plants receive. However, the Wikoda Sunflower is very short. It's very compact, but to the point where the system itself may be obstructed by other sources due to its low height.

With height comes increased access to unobstructed sunlight. Something like the Wikoda Sunflower that could be shortened down in some situations and then given an increased adjusted height when necessary would be the ideal solution.

One major application of sunlight, and one application that clashes with the shadows caused by the obstruction of light, is solar power. Solar panels produce a maximal amount of power when they have a somewhat direct line of sight with the sun. Too direct, and they may get damaged over time. Some countries have solar panels attached to heliostat-like systems that rotate the solar panel to face the sun directly [5]. However, solar panels are much heavier than basic mirrors. This implies that they require increased power or higher gear ratios, both of which cause them to be inefficient. Heavier masses at the end of a pitch arm mean more torque load. A higher torque load requires a higher torque input, which would require a stronger motor, which would have a higher power draw, which would cause the energy "profit" of the system to be far lower. Stanford also developed the AGILE system, an array of lenses that can be fitted to a solar panel to maximize the amount of light that is trapped and then transmitted to the solar panel. However, the AGILE system is dependent on the assumption that the solar panels are within the sun's path in the first place [2].

IV. Novelty:

A good majority of our design process involved taking a look at existing solutions, and then trying to combine their strengths whilst negating their weaknesses. Large heliostats are effective, but nobody could use one near their own home, so we knew our own design had to be smaller, just like the Wikoda Sunflower. However, the Wikoda Sunflower's low height gave it less access to unobstructed sunlight in general than the large heliostats. So, we made a design that could compress to the height of the Wikoda Sunflower, while also extending as high as we could physically make it. We worked to find situations where we could get the best of all worlds.

Systems that rotate solar panels to face the sun, while large, have output torques great enough to lift heavy solar panels. Our system's now as small as the Wikoda Sunflower, but the motors would have also shrunk down comparatively. This means that we lose out on the torque we would need to lift even a light mirror, an issue that we hoped to never run into. Needless to say, we ran into this issue after relying on tiny stepper motors, the kind that every Design and Tech student has the opportunity to work with. They are great for teaching students how to program due to their simplicity, but they are terrible at outputting high torques. The electromagnets used for stepping are just too small. In order to get the torque output of something like these solar panel movers, we had to use larger motors. However, there's only so big these motors can become before they throw the entire micro-heliostat off balance.

In order to create our own novel solution to the issue of lost torque, since this mirror has to rotate against gravity, we implemented a cycloidal drive. It's a gear system we have yet to see used for heliostats, yet it provides both precision and far more torque than is put in. The plan of combining the strengths of all these similar systems was what led us through the process of creating the micro-heliostat.

V. Impact:

The proposed micro-heliostat system has several potential applications related to solar energy utilization and sunlight accessibility. By dynamically redirecting sunlight throughout the day, the system could improve solar panel efficiency in environments where panels may not maintain an ideal orientation relative to the sun [2, 3]. Unlike stationary reflective systems, an actively tracked heliostat can continuously adjust its position to maximize usable light exposure.

The system could also reduce daytime dependence on artificial indoor lighting by redirecting natural sunlight into darker interiors or shaded areas of homes and buildings. This capability may improve energy efficiency while increasing the availability of natural lighting in residential settings. Additionally, the heliostat has the potential to improve sunlight access in shaded residential, suburban, or urban environments where buildings, trees, or other obstructions limit direct solar exposure.

More broadly, the project contributes to the development of compact, sustainable, and energy-efficient solar tracking solutions. By combining modular mechanical design, automated control systems, and real-time solar positioning, the system demonstrates how small-scale heliostat technology could be adapted for practical consumer or residential applications [4, 6].

VI. Methods:

Since this was a project with both a programming and physical component, we split up those portions. One group member handled the programming, and the other handled building and designing the system and frame. Both of us came together when working on the wiring and motors, as they required hardware knowledge for proper wiring and programming expertise to make them run.

To start this project, we needed to be sure that we could even program the motors we wanted. We researched microcontrollers and decided on the Raspberry Pi. We attempted to use a Raspberry Pi 1B at first, but it would not boot. No matter what OS we tried putting into it or what power supply we gave it, it would never work right. After switching it out, we learned that our first Raspberry Pi 1B had been broken, despite all of the websites and forums online that stated that it was nearly impossible for Raspberry Pis to fail because they are built to last. Yet, the same process for setup we used for the second one was the same.

Our OS of choice for this Raspberry Pi was the usual Linux-based Raspbian for this older Raspberry Pi. It seemed as if it would run the lightest, and we didn't need too many features for it to work. However, after working with it and attempting to make basic LED lighting programs, we had to accept a fact we were unwilling to. This was far too slow for the average user. People get slightly annoyed if their phone takes a few seconds longer to boot up, and this system was taking almost an entire half-second just for the cursor to register input. Furthermore, we did not have easy access to Ethernet ports anywhere in the classroom. We needed some connection to the internet in order to access the API that would give us the sun's location, that being the IPGeolocation API. The limitations of the Raspberry Pi 1 B were starting to become very apparent.

Having attempted to utilize the Raspberry Pi 1 B for two months, we still decided to switch to a newer model, and we ended up acquiring the Raspberry Pi 4. This board has wireless internet connectivity and Bluetooth, which allow us to remove some wires from the system for simplicity's sake. Setting up the newer Pi was extremely quick, and we completed the full setup process in less than one hour.

We knew that we wanted our system to have the ability to tilt the mirror up and down, to tilt the mirror left and right, and to move the mirror vertically and increase the height of the full apparatus. However, since these systems are going up against gravity, we knew for a fact that no logically-sized motor would provide enough torque alone. So, we needed to make use of methods by which we could increase the torque output. Considering that our system will have an entire 12 hours or so to cover the sky, we have no concern for any losses of speed.

Since this is a prototype system, we resolved to make the entire system modular. This means that all our parts must be easily swappable, yet hold their positions with relation to each other when in use. To support the design of these components, we used Autodesk Fusion 360. The CAD software allowed us to design, modify, and test the fitment of modular components before sending our components to our 3D printer, reducing the number of physical redesign cycles required during development. Since we were 3D printing this frame, we used the plasticity of PLA to our advantage. Where we wanted a seal or friction-based connection of some kind, we decreased the amount of error correction in the 3D model.

Starting off with the linear vertical movement, we decided upon a system that used a large screw and a screw acceptor to move the beam containing the acceptor up and down. The reason screws are so great at attaching objects is that they provide a mechanical advantage in that they provide for torque. You can't pull a screw out directly, but it's also really hard to push in a screw that is somewhat loosened. This means that the screw provides high-resistance linear motion and rotates easily as an input. It was the best method for rotational to linear movement we could think of.

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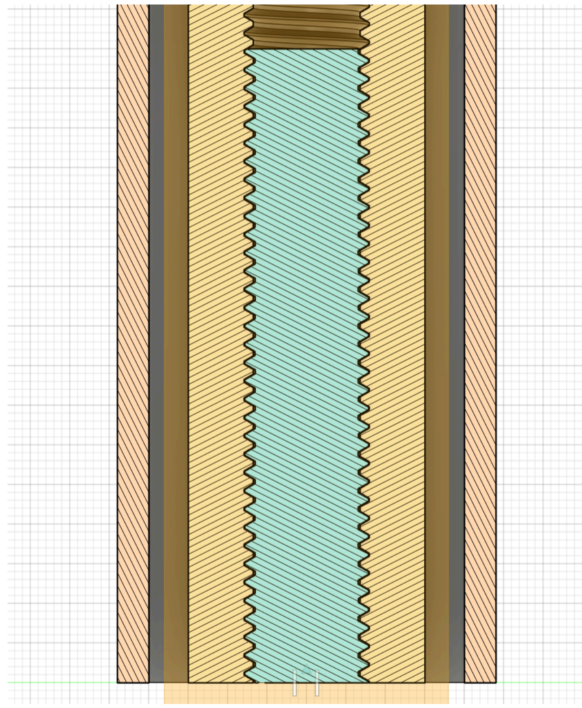

Figure 1: The linear actuator screw assembly cross-section

Figure 1 is a cross-section of our linear actuator system's screw setup. The blue portion is the screw, and the yellow portion is the beam with the threads that accept the screw. A single rotation of the screw while keeping the beam from rotating doesn't do much. Doing this over many, many rotations, however, lifts the beam and anything on top of it.

Despite the great torque output of this system, the stepper motors we were using at the time were struggling. Instead of rotating, they would either clip off the gears or start making clicking sounds as they attempted to skip over steps instead of actually stepping. We had to switch our motors to the industry standard NEMA-17 stepper motors. These are much larger, meaning larger coils, so a greater holding torque because magnetism increases as you add more coils. However, the addition of these larger and completely different steppers means that we had to go back on our current designs and then make them compatible with this new NEMA-17 stepper motor.

After deciding to use the NEMA-17 stepper motor, the design for the pitch (tilt up and down) and yaw (tilt left and right) directions necessitated a redesign, too. After learning how weak these older steppers were, we realized that we would need to ensure that torque stays high across the entire system, even in places where we believe we may not need it. We had initially planned to do this with a 1:4 gear ratio, but it was far too large to be viable. Good plastic gears usually have a bottom limit of 8 teeth for the smallest of gears, meaning that the smallest high-torque gear could be 32 teeth. Since we needed to account for printer issues, we needed to make that smallest gear even larger, so about 10 teeth. The conjugate would have to be 40 teeth. As we continued, it became clear that such a sub-optimal gear arrangement would not only be too large, but it would also require quite a few moving parts and full housings separate from each other.

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After further research, we discovered a minor “god” of torque. The system that was about to save our entire project is known as the cycloidal drive. The cycloidal drive has “cycloids” instead of gear teeth, which are these smooth bumps you would put on the perimeter of the gear, where the number of cycloids (not teeth anymore) is the torque ratio. This means that a single cycloidal gear with just eleven teeth gives us a 1:11 gear ratio, and in a smaller size than that single theoretical 1:4 gear ratio we were trying to make earlier.

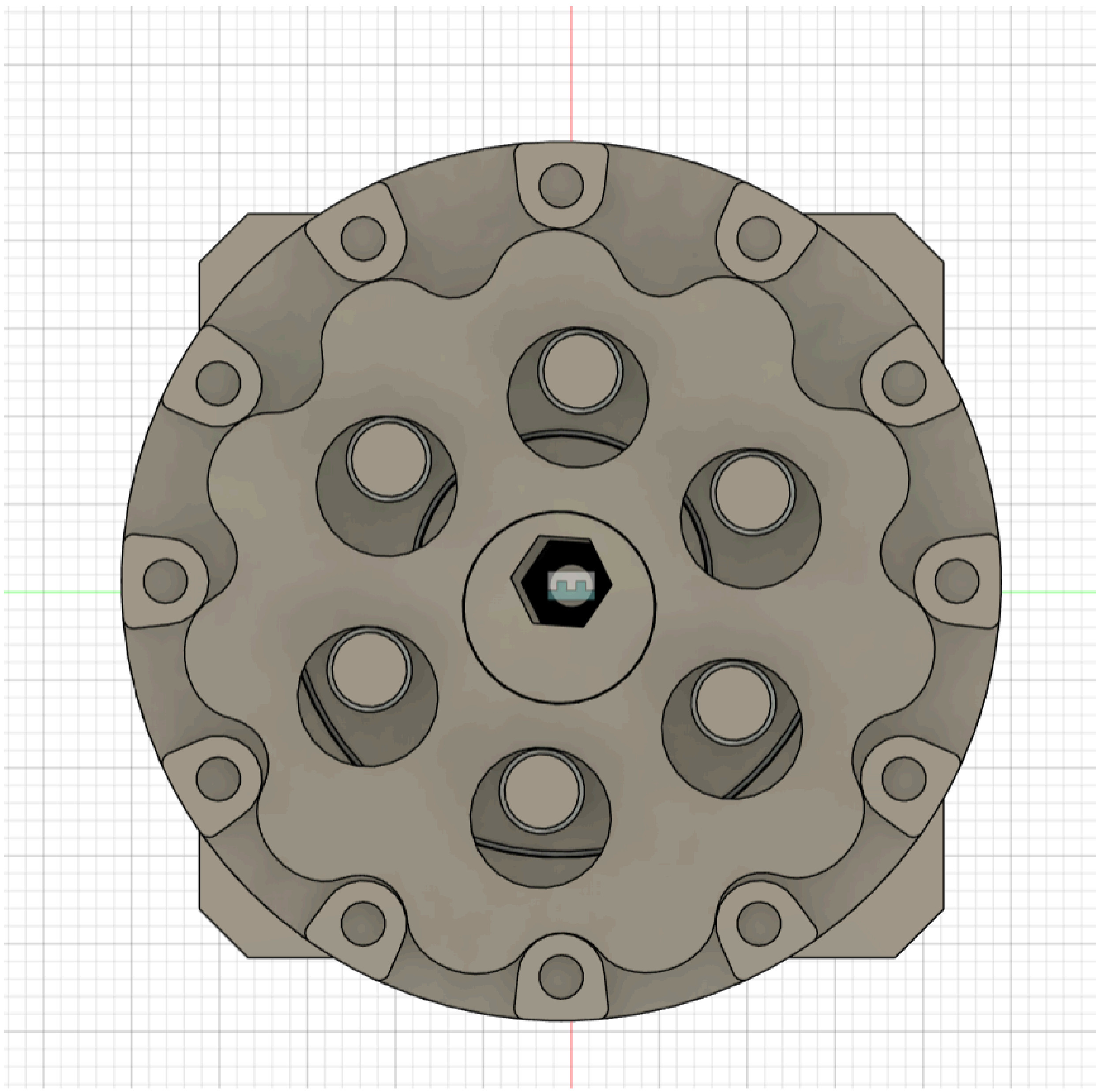

Figure 2: A top-down view of the cycloidal drive.

Figure 2 is an image of our cycloidal drive setup. The 11 bumps create a 1:11 gear ratio, meaning that our torque output is 11 times greater than our input while also making the output 11 times slower. I found this design online, thanks to PRUSA’s printables, but we naturally had to make small changes here and there for the drive to work with our own setup [1]. For starters, we

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found it necessary to shrink the model down… we shrunk it to about the size of a NEMA-17 stepper motor. Then, we had to account for specific tolerances. The design on the website has extremely tight pegs, which makes it difficult to remove the lid at this scale. We want to be able to remove it in case we feel as if we need to swap out cycloidal gears, allowing us to maintain modularity even in our most detailed and precise systems.

We maintained this modular design philosophy all the way through. We now needed to integrate these cycloidal drives into our pitch and yaw systems. Since we had believed that pitch would be the most difficult to test and prepare, we started with designing that system first. We used 3D modeled stepper motors as references so we could get a good idea of size and set our axles and housings up to accommodate these NEMA-17 stepper motors.

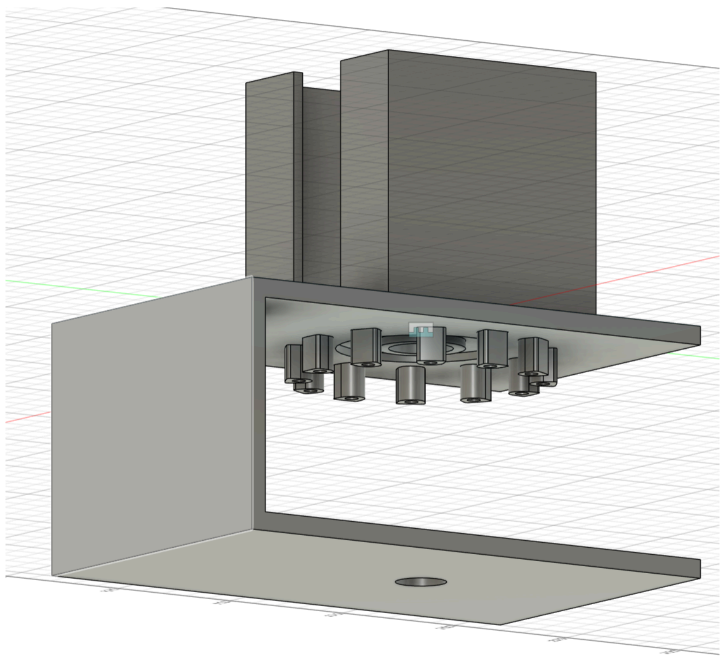

Figure 3: An early prototype of the pitch control system

Figure 3, shown above, is the early prototype that doesn't have the cycloidal gears in it in the image. Neither does it have the axles nor the steppers. However, it was designed using those reference files in order to ensure that they would fit together once printed. This early prototype lacks the modularity we wanted, but we later designed it once we knew how we wanted to combine the yaw system and the pitch system together. We were sure that this would be more than enough torque. Not only are NEMA-17s used in 3D printers, but they are also being given this high torque ratio in order to lift against gravity.

In designing the yaw system, we had to both figure out how to attach it to the linear actuator and how to attach it to the pitch module. Firstly, we would need another custom axle. Not one that could slide onto some kind of mirror arm like we sketched for the pitch axle, but rather one that provided support to an object sitting on top of it as it rotated left and right. Exactly how one's head would move when signaling a "no". We simply designed an extra plate on top of this axle and put four pegs on it, and then we could connect those four pegs into four holes in the pitch system that we added. We had also anticipated that we would use some kind of dovetail joint for connecting the linear actuator and the yaw setup, so we just made the other end of the dovetail for the frame of the yaw so we could slide it onto the existing linear actuator.

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The electrical control system consisted of a Raspberry Pi, two TMC2209 stepper motor drivers, two NEMA 17 stepper motors, a breadboard, and an external power supply. The external supply provided sufficient current for motor operation, as the Raspberry Pi alone could not safely power the stepper motors. Power from the supply was distributed through the breadboard rails, with each TMC2209 driver receiving dedicated power and ground connections. The Raspberry Pi GPIO pins were connected to the STEP, DIR, and ENABLE inputs of the drivers to control motor stepping, rotational direction, and driver activation. A shared ground connection between the Raspberry Pi, drivers, and power supply ensured reliable signal communication. Finally, the NEMA 17 motors were connected directly to the driver output terminals, enabling independent multi-axis motor control. Figure 4, shown below, is an image of our system completely wired up.

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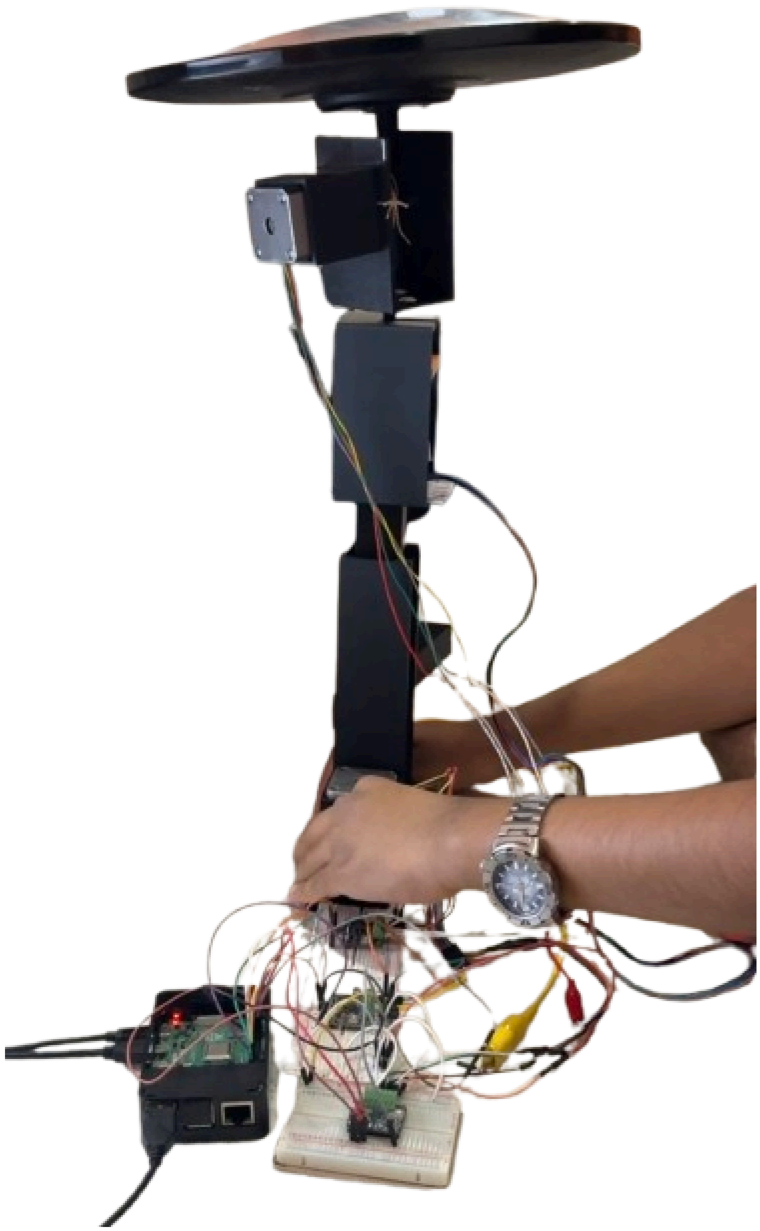

Figure 4: The entire heliostat, while running and fully assembled

Figure 5, depicted below, is a diagram of our system's physical and programmatic architecture. Red wires represent power, green wires represent signal, black wires represent USB/HDMI connections, and the dashed line represents the wifi connection between the IPGeolocation API and the Raspberry Pi.

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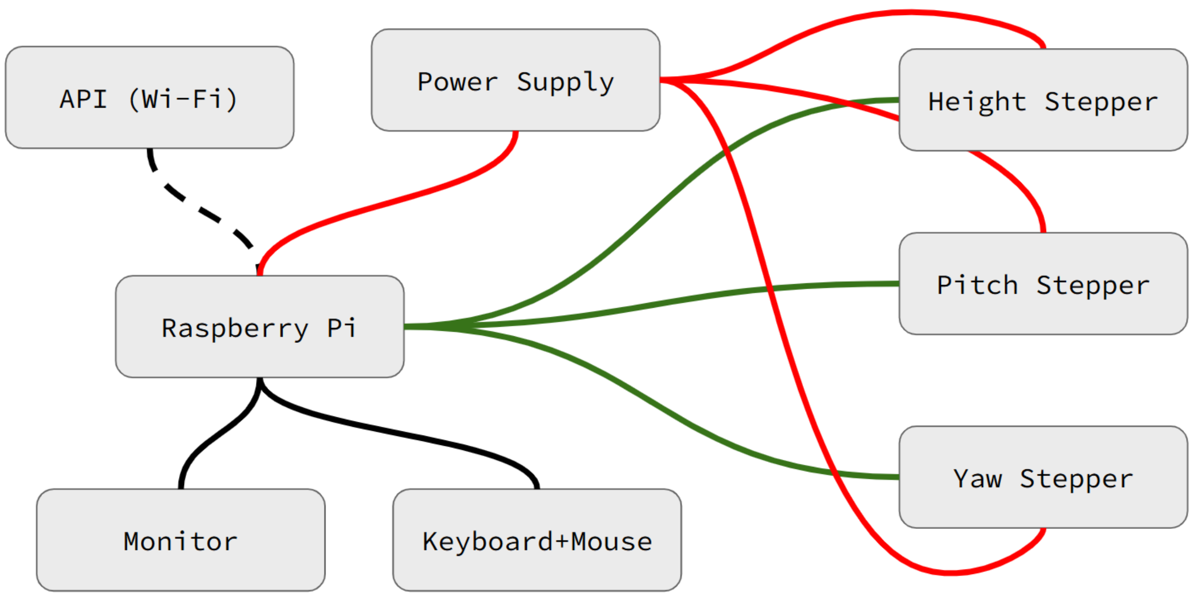

Figure 5: System physical and programmatical architecture diagram

With the way we've set this up, building the frame feels a lot like working with Legos. Everything either snaps or slides together, with minimal extra effort to keep those parts from sliding away from each other. While the keyboard, mouse, and monitor setup was found to be more on the difficult side for users, a system like this won't require all three to be connected at all times. They are only for initial adjustments and target positioning. The software, control logic, and implementation code used throughout this project are available in the project GitHub repository [7].

VII. **Results**:

Following multiple design iterations, prototyping cycles, and 3D-printed revisions, a functional heliostat configuration was successfully developed. The final design achieved a compact and modular structure, allowing the frame to be assembled in approximately five minutes while maintaining mechanical stability and ease of modification. The modular architecture also simplified component replacement and system adjustments during testing and development.

The incorporation of a 1:11 cycloidal drive significantly improved the torque capabilities of the stepper motor system. Without this gearing mechanism, the motors experienced a greater risk of missing steps when subjected to higher mechanical loads. By increasing torque output, the drive system enabled reliable rotational movement under demanding operating conditions. As a result, the heliostat successfully achieved its intended three modes of motion: pitch, yaw, and height adjustment.

Performance testing demonstrated that the system could move at speeds sufficient for solar tracking across an entire day while maintaining the ability to support mirrors positioned far from the system pivot point. This represents a significant mechanical challenge because increasing the distance between the pivot and the mirror increases the torque demand placed on

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the actuator system. Torque is defined as the product of applied force and radial distance from the pivot. In this application, the applied force is primarily gravitational, determined by the mirror mass multiplied by gravitational acceleration. Because the mirror's weight acts at a large distance from the pivot point, the resulting torque load becomes substantial. Despite these constraints, the system demonstrated the capability to lift and position high-moment assemblies effectively.

Additionally, the project established the mechanical and control foundation required for automated sunlight targeting. The team believes that, with sufficient solar position calculations derived from API data, the system could accurately redirect sunlight toward designated target locations. A polar-coordinate-based calculation framework was investigated to support this functionality; however, full implementation was not completed due to project time constraints.A demonstration of the system's functionality and operational behavior can be found in the project's accompanying YouTube video [8].

VIII. **Limitations**:

Several limitations affect the current implementation of the heliostat system. First, the reflective surface requires regular maintenance to preserve efficient light redirection. Environmental contaminants such as dust, dirt, moisture, and debris can accumulate on the mirror surface over time, reducing reflectivity and lowering overall system performance. This introduces a degree of user maintenance that may affect long-term practicality in outdoor deployments.

Additionally, effective operation depends heavily on installation conditions. The system requires placement in an elevated location with consistent access to direct sunlight to maximize tracking effectiveness and minimize obstruction from surrounding structures, vegetation, or terrain. These environmental constraints may limit deployment flexibility in densely built or heavily shaded areas.

The system also depends on a stable and continuous power supply to operate its motors, control electronics, and tracking mechanisms throughout the day. Because solar tracking requires sustained motion and computational control, power demand remains relatively constant during operation. As a result, relying exclusively on solar power may not provide sufficient reliability under cloudy weather conditions, reduced daylight availability, or extended periods of continuous use.

IX. **Conclusion**:

The heliostat system successfully demonstrates an automated multi-axis solar tracking platform built around a Raspberry Pi 4 controlling three stepper motors for height, pitch, and yaw adjustment. The integration of a 1:11 cycloidal drive gear ratio enables high-torque motion using compact stepper motors, improving the mechanical efficiency and load-handling capability of the design. Motor drivers, external power delivery, and Wi-Fi-based solar position data were integrated to enable continuous, accurate solar tracking without manual intervention.

Overall, the system achieves coordinated control of multiple degrees of freedom to maintain alignment with the sun throughout the day. This results in a compact, modular, and

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functional heliostat capable of consistent and precise solar orientation. The project demonstrates the successful integration of embedded control systems, mechanical motion design, and real-time positional data within a unified tracking platform. By enabling dynamic sunlight redirection, the system highlights the potential of automated heliostat technology for applications involving solar energy optimization, natural lighting enhancement, and targeted light delivery[6].

X. **Future Work**:

Future development of the heliostat system will focus on improving durability, environmental resilience, and user functionality. One proposed enhancement involves replacing selected plastic structural components with metal alternatives to increase mechanical strength, reduce wear, and improve long-term reliability under outdoor operating conditions. Additional protective features, such as an iris-based enclosure, are also being investigated to shield the reflective surface from dust accumulation, environmental exposure, and nighttime conditions, helping maintain consistent optical performance and reduce maintenance requirements.

Software improvements represent another area for continued development. Future iterations of the user interface may incorporate an optimization system that recommends optimal installation locations based on solar path calculations, environmental geometry, and sunlight availability. Such functionality could improve overall system performance by maximizing sunlight capture and reducing the need for manual placement experimentation. Together, these improvements would further enhance the practicality, robustness, and usability of the heliostat platform.

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